LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to DC systems: - power supply systems, three pin receptacle systems, schematic of an external power supply and over voltage pro
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to DC systems: - power supply systems, three pin receptacle systems, schematic of an external power supply and over voltage protection. (Multiple DC bus bar systems). Practical on 3-Pin Receptacle system.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on 3-Pin Receptacle system; Safety precautions: Verify power off before handling; Instructor supervision: Monitor connection integrity and voltage checks.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a DC power supply system?
(b) Identify the three pins in a standard three-pin receptacle system.
(c) Why is over-voltage protection necessary in avionics systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to DC power supply systems • Multiple DC bus bar configurations • Schematic of external power supply unit • Over voltage protection mechanisms	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of three pin receptacle systems • Pin identification and voltage ratings • Interconnection logic in DC distribution • Troubleshooting common receptacle faults	L & Ds	PPT, PDF	20 min
(c)	• Review of external power supply schematics • Integration of over voltage protection circuits • Bus bar system redundancy analysis • Summary of DC power distribution principles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the external power supply schematic ensures safe voltage delivery.
(b) Describe the role of over voltage protection in preventing avionics damage.
(c) Differentiate between the functions of the three pins in the receptacle system.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between DC and AC power systems to be discussed in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	AC systems - schematic of external power supply, auxiliary power units. Practical on 6-Pin Receptacle system. Introduction to Power Quality analysis (PQA)	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min